

Project Design Phase

Solution Architecture

Date	25 June 2025
Team ID	LTVIP2025TMID59612
Project Name	SB Foods - On-Demand Food Ordering Platform
Maximum Marks	4 Marks

Solution Architecture:

Solution Architecture bridges the gap between business needs and technical implementation. For the SB Foods application, the goal is to ensure a **scalable, responsive, and secure full-stack food ordering system** using the **MERN stack** (MongoDB, Express.js, React.js, Node.js), with clear data flow and modular components for users, restaurants, and administrators.

Objectives of the Architecture:

- Define how the application components interact across the tech stack.
- Ensure clear separation of concerns: UI, business logic, data storage.
- Enable smooth data flow between client, server, and database.
- Support future scaling (e.g., more users, multi-region deployment).
- Allow role-based access (User, Restaurant, Admin).

Architecture Layers and Components:

Layer	Components / Tools Used	Description
Frontend	React.js, HTML/CSS, Axios	User interface for all roles (User, Restaurant, Admin). Components include Login, Cart, Orders, Dashboard, etc.
Backend	Node.js, Express.js	RESTful API server that handles routing, authentication, product management, order processing.
Database	MongoDB (with Mongoose ODM)	Stores all structured data: Users, Restaurants, Products, Orders, Carts, Admin data.
Authentication	JWT (JSON Web Tokens) + bcrypt	Secure login for users, restaurants, and admin roles.
Hosting	Localhost (Dev) / Future: Vercel/Heroku/Render	Deployment of frontend and backend in cloud or container environments.
Dev Tools	Git, VS Code, MongoDB Atlas, Postman	Development and testing tools used for building and verifying the application.

Data Flow Summary

1. User Journey:

- User registers/login → token stored → fetches food listings → adds to cart → places order → order stored in DB.

2. Restaurant Journey:

- Logs in → lists or edits products → products stored in DB → can view orders made on their items.

3. Admin Journey:

- Logs in → views all users/products/orders → approves restaurants/products → manages categories/promotions.

Example - Solution Architecture Diagram:

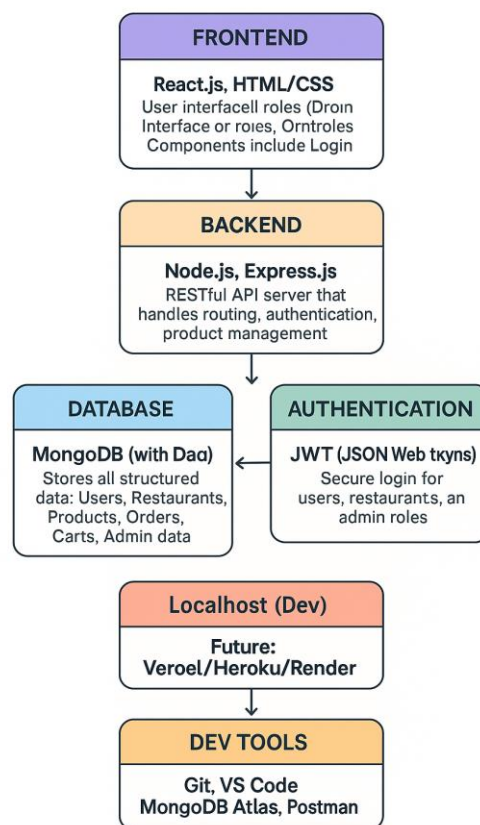


Figure 1: Architecture and data flow of the food ordering sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Features Defined in Architecture

Feature	Handled By	Status
User Authentication	JWT, bcrypt, Express.js	✓ Implemented
Product Listings	MongoDB, React, Axios	✓ Implemented
Cart & Order Management	React, Express, MongoDB	✓ Implemented
Role-Based Access Control	JWT Middleware	✓ Implemented
Admin Approval Flow	Admin Panel, DB updates	✓ Implemented
API Layer Security	Auth middleware, CORS	✓ Implemented
Scalability Provision	Modular Code, REST APIs	Scalable
Database Optimization	Mongoose ODM	✓ Optimized

Notes

- The current design supports future enhancements like third-party payment integration, push notifications, and analytics.
- With MongoDB Atlas, the platform can be scaled vertically and horizontally based on usage.
- Role-based segregation ensures minimal coupling and better maintainability.